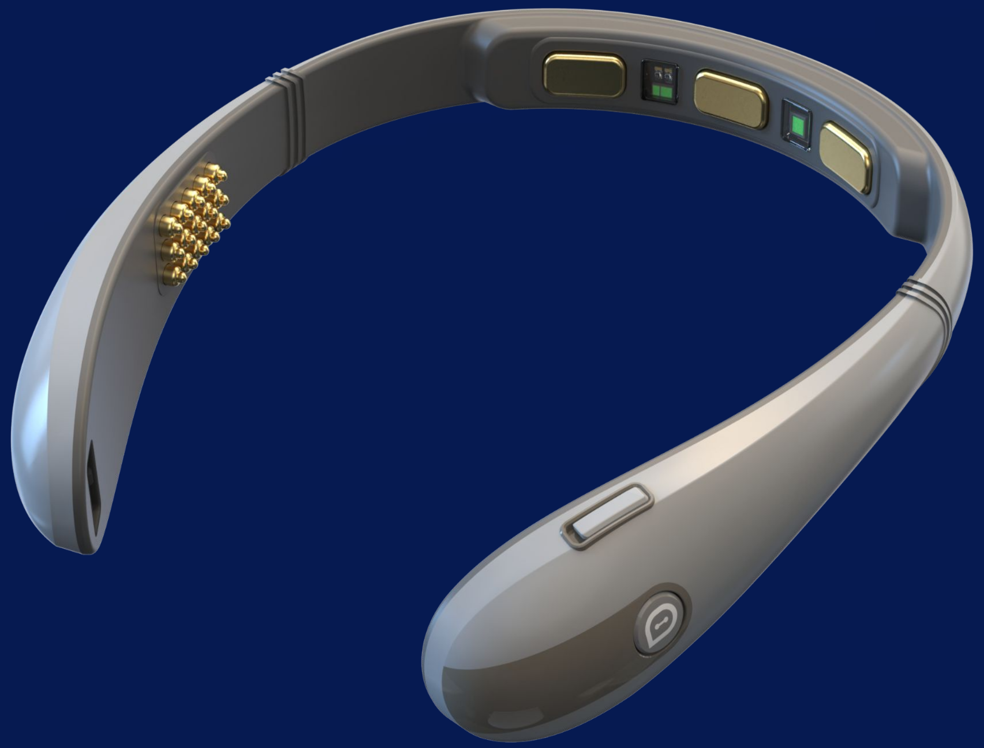
Brain-Life
Understand your brain - Master your life

Neuro-AI for Smart Healthcare: From Early Screening to Personalized Intervention and Remote Care



SiHUB event 31 July 26
Dr. Xuan Tran
Chairman, Brain-Life Link Technology JSC.

Digital + preventive healthcare signals

Recent news shows Vietnam is building the data backbone and prevention mindset that Neuro-AI can extend.

NEWS 01

HCMC: lifelong EHRs for ~15M people

Why it matters for healthcare partners

- Longitudinal data instead of one-off visits
- Foundation for early risk detection and follow-up
- Natural entry point for community screening pilots

NEWS 02

Clinical voices recognize mind–body prevention

Relevant points from Tuổi Trẻ / Chợ Rẫy

- Meditation supports, not replaces, medical care
- Stress links to cardiovascular risk and adherence
- Prevention needs feedback, guidance and continuity

BRIDGE TO BRAIN-LIFE

If health records create the longitudinal backbone, Neuro-AI wearables add the missing daily layer:
cognitive fatigue, stress, recovery, attention and brain-health trends.

VnExpress: https://vnexpress.net/15-trieu-dan-tp-hcm-duoc-lap-ho-so-suc-khoe-dien-tu-theo-vong-doi-5091450.html
Tuổi Trẻ: https://tuoitre.vn/bac-si-benh-vien-cho-ray-noi-ve-tac-dung-cua-thien-20251226140718647.htm

Vietnam healthcare is approaching an inflection point

Three forces are converging: chronic disease, rapid ageing and a national push for data-driven healthcare.

~80%

of deaths in Vietnam are caused by noncommunicable diseases

WHO Viet Nam, 2025

>20%

of the population will be aged 60+ by 2036

UNFPA Viet Nam, 2026

2025–2030

Vietnam's health digital strategy targets connected, standardized, real-time data

Ministry of Health, 2025

The central question

How can Vietnam move from a system that mainly reacts to illness to one that identifies risk early, supports prevention and delivers continuous care closer to people?

Sources: WHO Viet Nam — NCDs; UNFPA Viet Nam — Population ageing; Viet Nam Ministry of Health — Digital transformation strategy 2025–2030

From episodic treatment to proactive, continuous, lifelong care

Digitalization matters only when it changes decisions, access and outcomes.

Today: episodic and hospital-centred

- Symptoms appear
- Hospital visit
- Single-point assessment
- Treatment after deterioration

High downstream cost • limited visibility between visits

Next: preventive, distributed and continuous

- Continuous sensing and self-awareness
- Early risk identification
- Timely, personalized intervention
- Longitudinal follow-up at home and community

Earlier action • greater reach • measurable outcomes

Most healthcare decisions rely on snapshots. Health risks develop continuously.

Concept aligned with WHO primary healthcare and Viet Nam's 2025–2030 digital health strategy.

Brain-Life

Brain health is a life-course issue — not only a specialist issue

The same sensing and data infrastructure can support different needs at different stages of life.

Development

Neurodevelopment, autism support, learning and engagement

Daily well-being

Focus, stress, fatigue, recovery and healthy habits

Safety & work

Cognitive load, fatigue, errors and occupational health

Healthy ageing

Cognitive screening, independence and caregiver support

Clinical pathways

Stroke, migraine, dementia and Parkinson's research

One Neuro-AI platform vision: wellness today → research and validated clinical applications tomorrow

WHO: neurological conditions affect more than 1 in 3 people worldwide and are the leading cause of illness and disability.

Brain-Life as an assisted data platform

Brain-Life is a Neuro-AI assisted-data platform that adds longitudinal brain and physiological signals to existing healthcare workflows.

1

SENSE

EEG + fNIRS + PPG + behaviour

2

UNDERSTAND

Focus, stress, fatigue, engagement, load

3

ACT

Neurofeedback, breathing, audio and training

4

LEARN

Personalization and longitudinal model improvement

Output: Risk indicators • longitudinal trends • personalized intervention • early warnings / referral support

Focus+ for a healthier, happier and more productive society

Prevention begins before disease: helping people recognize strain, recover earlier and build healthier cognitive habits.

Population-level preventive value

Self-awareness

See focus, stress and fatigue patterns rather than relying only on subjective feeling.

Earlier recovery

Use short breathing, meditation, audio or rest interventions before overload accumulates.

Better daily performance

Reduce avoidable errors and improve consistency in learning and knowledge work.

Potential system value

Lower downstream burden is a health-economic hypothesis to validate in real deployments.

Early Brain-Life user evidence

75%
of 200 users improved focus

43%
fewer task errors after meditation

32%
fewer task errors with binaural beats

7-day pilot — 13 users (10-point scale)

−1.36
stress

+2.08
focus

−2.00
body tension

Early evidence — not a clinical outcome claim. Larger controlled studies are needed.

Brain-Life internal studies reported in the June 25, 2026 pitch deck: n=200 and n=13.

USE CASE 2

Neurodevelopment and mental well-being

Objective signals can complement — not replace — clinical observation, validated scales and professional judgement.

Autism & neurodevelopment

- Support research on attention, engagement and sensory response
- Short, child-friendly tasks and games
- Track response over time
- Potential aid for referral prioritization — not standalone diagnosis

Mental well-being

- Stress and cognitive-fatigue monitoring
- Personalized relaxation and recovery
- Support for students, workers and caregivers
- Escalation to human support when risk is identified

Partnership study design

- Clinical lead defines intended use and outcomes
- Compare with validated assessments
- Consent, safeguarding and data governance
- Evaluate usability, equity and clinical utility

Clinical boundary: current Brain-Life products are not approved to diagnose autism or mental disorders.

WHO supports integration of mental health into primary healthcare; autism services should be person- and family-centred.

USE CASE 3

Healthy ageing and longevity: detect decline earlier, preserve independence longer

Vietnam’s rapid ageing makes scalable cognitive and functional monitoring a public-health priority.

2036

Viet Nam becomes an “aged” society

More than 1 in 5 people aged 60+

Healthy ageing means maintaining functional ability, not merely extending lifespan.

A practical community-to-hospital workflow

10-15 min screen
Digital task + cognitive questionnaire + brain/physiology signals

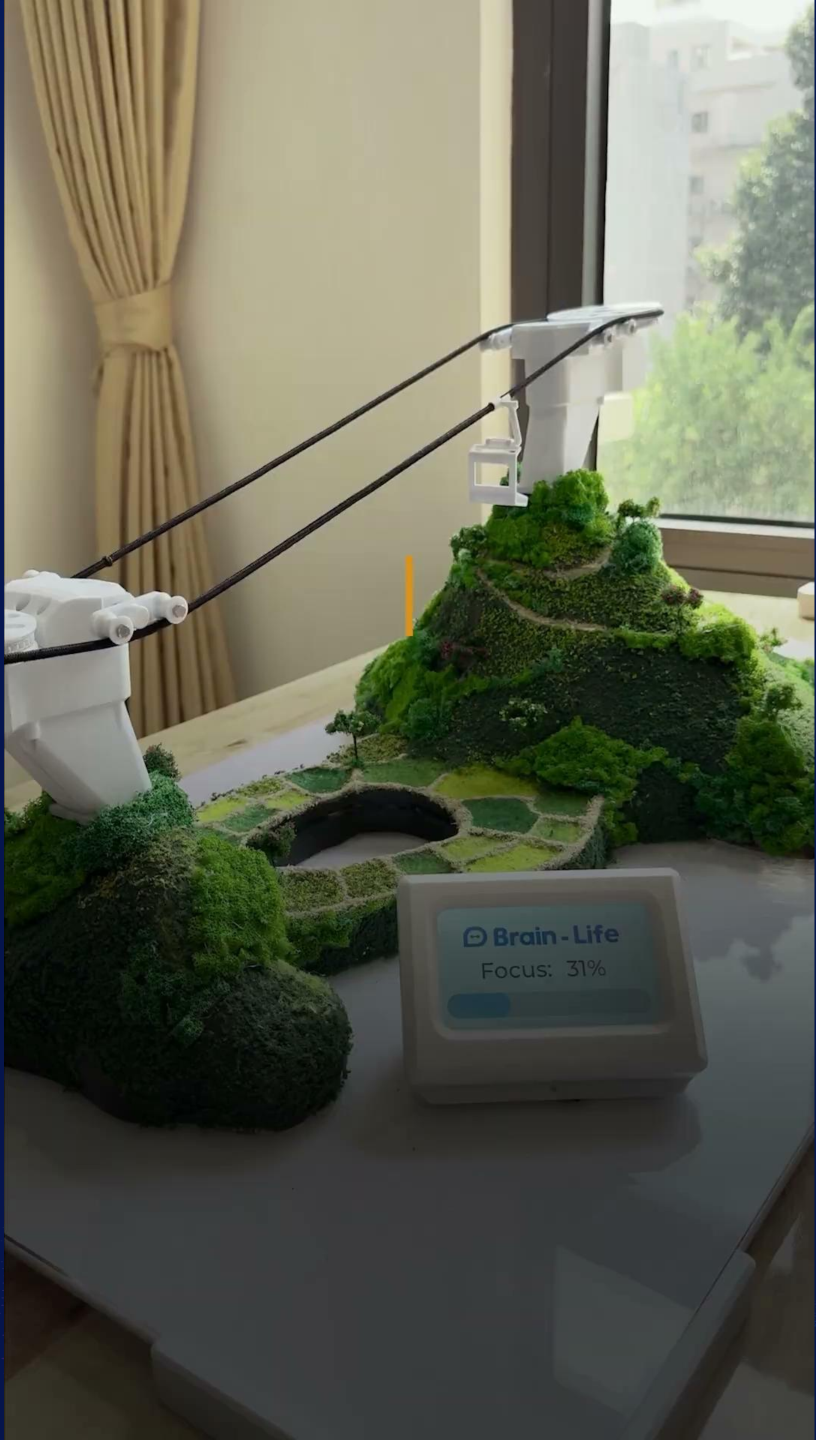
→ **Risk stratification**
Identify patterns that merit deeper professional assessment

→ **Referral**
Clinical review, diagnosis and care planning where appropriate

→ **Follow-up**
Repeated measurements reveal trends rather than one snapshot

Goal: prioritize scarce specialist capacity while supporting people to remain independent at home.

UNFPA Viet Nam, 2026; WHO Integrated Care for Older People (ICOPE).



USE CASE 4

Care robots as a human-care multiplier

The robot is the interface; wearables and clinical systems provide data; human professionals remain accountable.

Older person

Daily interaction and support

Hospital

Clinical oversight and escalation

CARE ROBOT

conversation • reminders • guided exercises • telepresence • escalation

Family & caregivers

Updates, calls and shared care plans

Community health

Screening and local follow-up

Focus+ / sensors → secure data platform → alerts and personalized support

WHO: assistive technology can support cognition, independence and reduce pressure on carers; AI must be co-designed with older people and avoid age bias.

USE CASE 5

Brain-related conditions: continuous monitoring across the care journey

The opportunity is not to diagnose everything from a headband, but to capture useful longitudinal signals between clinical visits.

Stroke

Recovery engagement, fatigue, attention and home follow-up after discharge

Near-term pilot

Migraine / headache

Track triggers, stress, sleep context and symptom patterns over time

Research use

Alzheimer's / dementia

Cognitive trend monitoring, caregiver support and referral pathways

Longitudinal study

Parkinson's

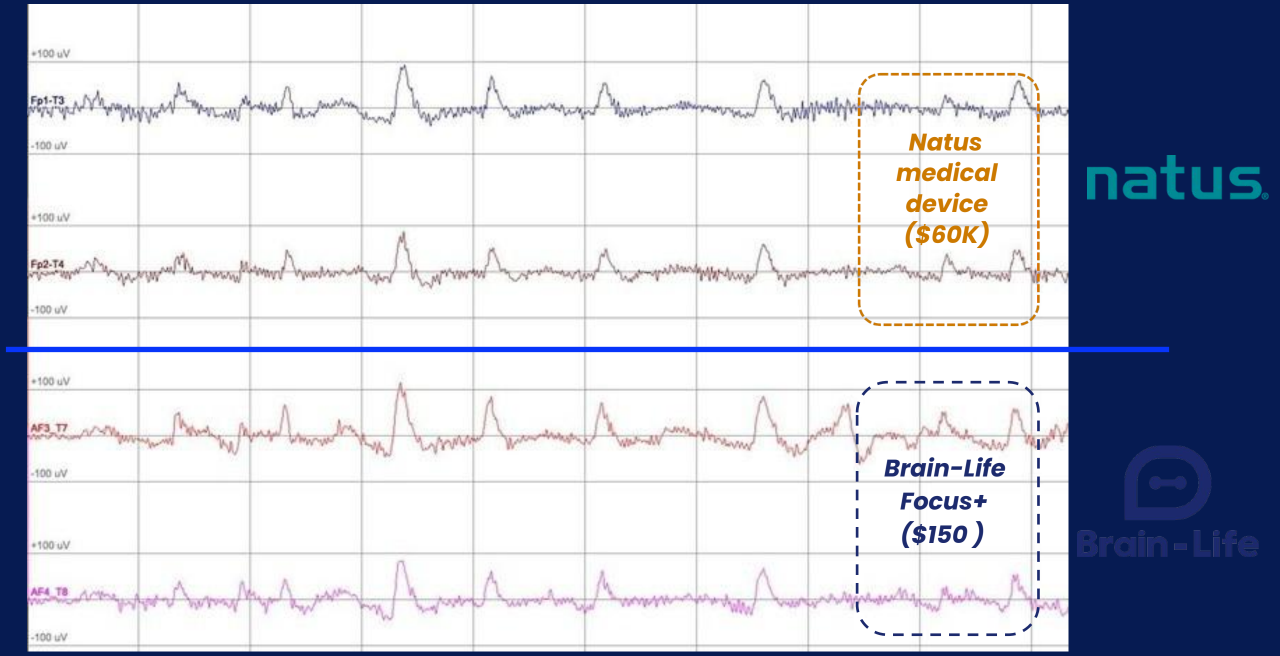
Combine cognitive, physiological and movement data for monitoring research

Multimodal study

Current status: these are research and future clinical directions. Each intended use requires disease-specific evidence, clinician oversight and regulatory review.

WHO: stroke is the largest contributor to neurological health loss; dementia affects 57M people globally; Parkinson's prevalence has doubled in 25 years.

Signal validation - compared with Natus medical device



Time series EEG signal Comparison

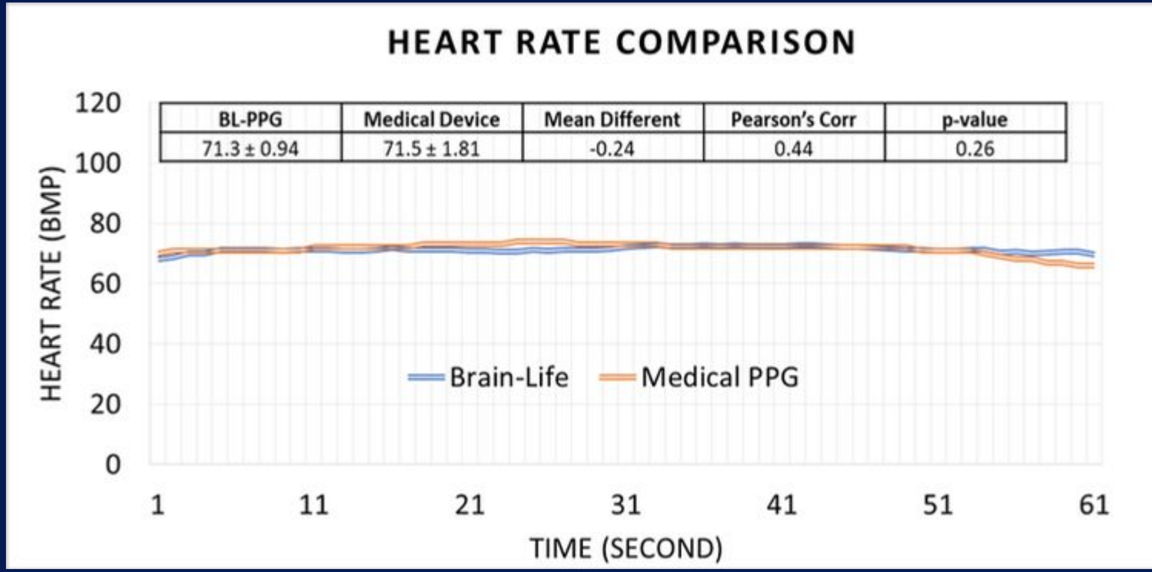
"The graphs showed that Focus+ was able to capture similar patterns to those of Natus in all three brain states."(*)

In a test of relaxation, free rest, and focus states, measuring at the same time on the same person, using Brain-Life's device, and in comparison, with EEG signals from the Natus machine used in clinical diagnostics.

— MD. MSc. Bui Diem Khue

University of Medicine & Pharmacy HCMC (UMP) Deputy General Secretary of the Vietnam Society of Sleep Medicine

(*) Reference: https://vnexpress.net/nguyen-mau-thiet-bi-theo-doi-nao-than-kinh-made-in-vietnam-4916383.html



BL-PPG	Medical Device	Mean Different	Pearson's Corr	p-value
71.3 ± 0.94	71.5 ± 1.81	-0.24	0.44	0.26

PPG Comparison
(Brain-Life vs Medical Device)

Heart rate signals measured by Brain-Life's wearable device closely match those from a medical-grade reference system over a 60-second period. There is **no significant difference between the two signals**. This indicates our device is clinically reliable.

PARTNERSHIP AGENDA

Four projects that can start small and generate decision-ready evidence

Each project should be co-owned by a clinical lead, a health-system sponsor and Brain-Life.

1. Older-adult screening

3 sites • ~200 older adults • 10-15 minute task • referral feasibility

Outcome: usability, signal quality and correlation with established screens

2. Neurodevelopment & well-being

Child-friendly or mental well-being protocol with validated scales

Outcome: engagement markers, workflow fit and ethical safeguards

3. Post-stroke follow-up

Home/community monitoring after discharge with rehabilitation team

Outcome: adherence, fatigue/engagement trends and escalation workflow

4. Care robot integration

Connect wearable data, reminders, telepresence and caregiver alerts

Outcome: acceptance, caregiver burden and service-efficiency measures

Suggested scopes are designed as feasibility projects, not medical efficacy trials.

Thank you for your attention !

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